

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

*I _____Chalamala Sai Harshitha (192424346)____certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C. Sai Harshitha

Annexure A

Short Answer Test

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Given:

- Study hours, $x_1 = 2$, Attendance, $x_2 = 1$
- Weight, $w_1 = 0.6$, Weight, $w_2 = 0.4$
- Bias, $b = -0.5$

Step 1: Net Input

$$\text{Net Input} = \sum_{i=1}^n w_i x_i + b$$

For two inputs,

$$\begin{aligned}\text{Net Input} &= (w_1 x_1) + (w_2 x_2) + b \\ &= (0.6 \times 2) + (0.4 \times 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 \\ &= 1.1\end{aligned}$$

Step 2: Step Activation Function

$$y = \begin{cases} 1, & \text{if Net Input} \geq 0 \\ 0, & \text{if Net Input} < 0 \end{cases}$$

Since,

$$\begin{aligned}1.1 &\geq 0 \\ y &= 1\end{aligned}$$

The perceptron output is **1**, indicating that the **student passes**.

Final Answer

- **Net Input = 1.1**
- **Output = 1**
- **Student Passes**

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Backpropagation (Delta Learning Rule)

Given:

- Initial Weight, $w = 0.5$
- Learning Rate, $\eta = 0.1$, Input, $x = 2$
- Actual Output, $y = 0.6$, Target Output, $t = 1$

Step 1: Error

$$\begin{aligned}\text{Error} &= t - y \\ \text{Error} &= 1 - 0.6 = 0.4\end{aligned}$$

Step 2: Weight Update (Delta Rule)

$$\begin{aligned}\Delta w &= \eta \times (t - y) \times x \\ \Delta w &= 0.1 \times 0.4 \times 2 \\ &= 0.08\end{aligned}$$

Step 3: Updated Weight

$$\begin{aligned}w_{\text{new}} &= w_{\text{old}} + \Delta w \\ w_{\text{new}} &= 0.5 + 0.08 \\ &= 0.58\end{aligned}$$

Conclusion

After one iteration of training using the **delta learning rule**, the weight is updated from **0.5** to **0.58**, reducing the prediction error.

Final Answer

- **Error = 0.4**
- **Weight Update (Δw) = 0.08**
- **Updated Weight = 0.58**

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Given:

- True Positives (TP) = 180
- True Negatives (TN) = 150
- False Positives (FP) = 20
- False Negatives (FN) = 50

Accuracy

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ &= \frac{180 + 150}{180 + 150 + 20 + 50} \\ &= \frac{330}{400} \\ &= 0.825 = 82.5\%\end{aligned}$$

Precision

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{180}{180 + 20} \\ &= \frac{180}{200} \\ &= 0.9 = 90\%\end{aligned}$$

Sensitivity (Recall)

$$\begin{aligned}\text{Sensitivity (Recall)} &= \frac{TP}{TP + FN} \\ &= \frac{180}{180 + 50} \\ &= \frac{180}{230} \\ &= 0.7826 = 78.26\%\end{aligned}$$

Specificity

$$\begin{aligned}\text{Specificity} &= \frac{TN}{TN + FP} \\ &= \frac{150}{150 + 20} \\ &= \frac{150}{170} \\ &= 0.8824 = 88.24\%\end{aligned}$$

Conclusion

The model has **82.5% accuracy**, **90% precision**, **78.26% sensitivity**, and **88.24% specificity**. It performs well overall and correctly predicts most cases. However, the sensitivity is lower, which means some positive cases are missed. Therefore, improving sensitivity would make the model better for medical diagnosis, where detecting all positive cases is important.

Accuracy = 82.5%

Precision = 90%

Sensitivity (Recall) = 78.26%

Specificity = 88.24%

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Genetic Algorithm

Given:

- Fitness Function: $f(x) = x^2$
- Initial Population: {2, 4, 6, 8}

Step 1: Fitness Function

$$f(x) = x^2$$

Calculate Fitness Values

Individual (x)	Fitness $f(x) = x^2$
2	$2^2 = 4$
4	$4^2 = 16$
6	$6^2 = 36$
8	$8^2 = 64$

Step 2: Select the Best Candidates

The individuals with the highest fitness values are:

- 8 → Fitness = 64
- 6 → Fitness = 36

These are selected as the parent chromosomes.

Step 3: Perform Crossover

Binary representation:

- 8 = 1000
- 6 = 0110

Exchange the **least significant bit (LSB)**:

- Parent 1: **1000**

- Parent 2: **0110**

After swapping the last bit:

- Offspring 1: **1000** (8)
- Offspring 2: **0110** (6)

Offspring: 8 and 6

Step 4: Role of Mutation

Mutation randomly changes one or more bits in a chromosome. It helps:

- Increase diversity in the population.
- Prevent the algorithm from getting stuck in a local optimum.
- Improve the chances of finding the best solution.

Final Answer

- **Fitness Values:** 4, 16, 36, 64
- **Selected Candidates:** 8 and 6
- **Offspring after Crossover:** 8 and 6

5. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration.

Backpropagation Using Delta Learning Rule

Given:

- Initial Weight, $w = 0.4$
- Learning Rate, $\eta = 0.2$, Input, $x = 3$
- Actual Output, $y = 0.5$, Target Output, $t = 0.9$

Step 1: Error

$$\begin{aligned}\text{Error} &= t - y \\ \text{Error} &= 0.9 - 0.5 = 0.4\end{aligned}$$

Step 2: Weight Update (Delta Rule)

$$\begin{aligned}\Delta w &= \eta \times (t - y) \times x \\ \Delta w &= 0.2 \times 0.4 \times 3 \\ &= 0.24\end{aligned}$$

Step 3: Updated Weight

$$\begin{aligned}w_{\text{new}} &= w_{\text{old}} + \Delta w \\ w_{\text{new}} &= 0.4 + 0.24 \\ &= 0.64\end{aligned}$$

Conclusion

After one iteration of training using the **delta learning rule**, the weight is updated from **0.4** to **0.64**, which helps reduce the prediction error.

Final Answer

- **Error = 0.4**
- **Weight Update (Δw) = 0.24**
- **Updated Weight = 0.64**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand